AQA 3.1.9 Rate equations

AQA A-level Chemistry · Revision questions · Sheet 4 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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19 marks · about 25 minutes

Key information

- rate = $k[A]^m[B]^n$ — orders come from experimental data, not from the balanced equation
- Doubling a concentration: rate × 1 (zero order), × 2 (first order), × 4 (second order)
- Units of k: rearrange k = rate ÷ (concentration terms), with rate in mol dm^{-3} s^{-1}, and cancel
- ln k = $-E_a/RT$ + ln A, so the gradient of ln k against 1/T = $-E_a/R$
- Use E_a in J mol^{-1} and T in K with R = 8.31 J K^{-1} mol^{-1}; convert to kJ mol^{-1} at the end

Section A: Short-answer questions

01 A graph of ln k (y-axis) against 1/T (x-axis) for a reaction is a straight line with a gradient of −6500 K. Calculate the activation energy, in kJ mol^{-1}, for this reaction. The gas constant R = 8.31 J K^{-1} mol^{-1}

[2 marks]

02 A graph of ln k against 1/T has an intercept on the y-axis of 28.0. Calculate the value of the Arrhenius constant, A.

[1 mark]

03 The rate equation for a reaction is rate = k[A][B]. Deduce the units of the rate constant, k.

[1 mark]

04 For a reaction, A = 1.5×10^{11} s^{-1} and E_a = 75.0 kJ mol^{-1}. Use ln k = ln A − E_a/RT to calculate k at 350 K. R = 8.31 J K^{-1} mol^{-1}

[2 marks]

05 The rate equation for a reaction is rate = $k[A]^2$. The rate constant is 0.36 mol^{-1} dm^3 s^{-1}. Calculate the concentration of A when the rate is 2.25×10^{-3} mol dm^{-3} s^{-1}.

[2 marks]

Section B: Longer questions

06 Substances P and Q react in solution at a constant temperature.

The initial rate of reaction was studied in three experiments by measuring the change in concentration of P over the first five seconds of the reaction. The data obtained are shown in Table 1.

Experiment	Time after mixing / s	[P] / mol dm^{-3}	[Q] / mol dm^{-3}
1	0	1.00×10^{-2}	1.25×10^{-2}
	5.0	0.92×10^{-2}	not measured
2	0	2.00×10^{-2}	1.25×10^{-2}
	5.0	1.84×10^{-2}	not measured
3	0	0.50×10^{-2}	2.50×10^{-2}
	5.0	0.34×10^{-2}	not measured

06.1 The initial rate of reaction of P in Experiment 1 is 1.6×10^{-4} mol dm^{-3} s^{-1}. Calculate the initial rate of reaction of P in Experiments 2 and 3.

[1 mark]

06.2 Determine the order of reaction with respect to P and the order of reaction with respect to Q.

[2 marks]

06.3 A reaction between substances R and S was second order with respect to R and second order with respect to S.

At a given temperature, the initial rate of reaction was 1.20×10^{-3} mol dm^{-3} s^{-1} when the initial concentration of R was 1.00×10^{-2} mol dm^{-3} and the initial concentration of S was 2.45×10^{-2} mol dm^{-3}

Calculate a value for the rate constant, k, for the reaction at this temperature. Give the units for k.

[3 marks]

07 Propanone reacts with bromine in alkaline conditions.

$CH_3COCH_3 + Br_2 + OH^- \rightarrow CH_3COCH_2Br + Br^- + H_2O$

The rate equation for this reaction is rate = $k[CH_3COCH_3][OH^-]$

Table 1 shows the initial rate of this reaction for experiments using different mixtures containing propanone, bromine and hydroxide ions.

Experiment	$[CH_3COCH_3]$ / mol dm^{-3}	$[Br_2]$ / mol dm^{-3}	$[OH^-]$ / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
1	1.50×10^{-2}	2.50×10^{-2}	2.50×10^{-2}	2.75×10^{-11}
2	1.50×10^{-2}	2.50×10^{-2}		8.25×10^{-11}
3	3.75×10^{-3}	5.00×10^{-2}	1.00×10^{-1}	

Complete Table 1. Use the data from experiment 1 to calculate the rate constant k for this reaction. Give the units for the rate constant.

[5 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	M1 E_a = –gradient × R = 6500 × 8.31 (= 54 015 J mol^{-1}) M2 E_a = 54.0 kJ mol^{-1}	*Source: Practice question (written for this set)*	2
02	A = $e^{28.0}$ = 1.45 × 10^{12}	*Source: Practice question (written for this set)*	1
03	mol^{-1} dm^3 s^{-1}	*Source: Practice question (written for this set)*	1
04	M1 ln k = ln(1.5 × 10^{11}) – 75 000 ÷ (8.31 × 350) = –0.053 M2 k = 0.95 s^{-1}	*Source: Practice question (written for this set)* Allow 0.948–0.95	2
05	M1 $[A]^2$ = 2.25 × 10^{-3} ÷ 0.36 (= 6.25 × 10^{-3}) M2 [A] = 0.0791 mol dm^{-3}	*Source: Practice question (written for this set)* Allow 0.079	2
06.1	Experiment 2 = 3.2 × 10^{-4} **and** Experiment 3 = 3.2 × 10^{-4} (mol dm^{-3} s^{-1})	*Source: 7405/2 Jun 2019 Q04* Both needed. Original asks students to complete a table	1
06.2	P order = 1 Q order = 2		2
06.3	M1 k = rate ÷ $[R]^2[S]^2$ = 1.20 × 10^{-3} ÷ (1.00 × 10^{-2})^2(2.45 × 10^{-2})^2 M2 k = 2.00 × 10^4 M3 units mol^{-3} dm^9 s^{-1}	M2 allow 1.99 × 10^4 Allow consequential units for their expression in M1	3
07	M1 Experiment 2 $[OH^-]$ = 7.50 × 10^{-2} M2 Experiment 3 rate = 2.75 × 10^{-11} M3 k = rate ÷ $[CH_3COCH_3][OH^-]$ = 2.75 × 10^{-11} ÷ (1.5 × 10^{-2} × 2.5 × 10^{-2}) M4 k = 7.3(3) × 10^{-8} M5 units mol^{-1} dm^3 s^{-1}	*Source: 7405/2 Jun 2024 Q01.2* If rearrangement upside down lose M3 but M4 = 1.36 × 10^7 as ECF M5 allow mol dm^{-3} s as ECF	5